

Intermediate Elective 2

Abby Malech

2026-05-15

Set-up

Read-in packages

```
# general use and cleaning
library(tidyverse)
library(here)
library(janitor)
library(snakecase)

# customization
library(prismatic)
library(stringr)

# wrapping axis labels
library(scales)

# reading in .xlsx files
library(readxl)

# arranging plots
library(patchwork)

## 1. LOAD PACKAGES & SETUP ----
suppressPackageStartupMessages({
  if (!require("pacman")) install.packages("pacman")
  pacman::p_load(
    tidyverse, # Easily Install and Load the 'Tidyverse'
    ggtext,    # Improved Text Rendering Support for 'ggplot2'
```

```

    showtext,    # Using Fonts More Easily in R Graphs
    janitor,     # Simple Tools for Examining and Cleaning Dirty Data
    scales,      # Scale Functions for Visualization
    glue,        # Interpreted String Literals
    gghighlight # Highlight Lines and Points in 'ggplot2'
  )))

### figure size ----
camcorder::gg_record(
  dir      = here::here("temp_plots"),
  device   = "png",
  width    = 16,
  height   = 20,
  units    = "in",
  dpi      = 320
)

```

Read-in data

```

# WEEK 4 data

# vegetation survey data
veg <- read_csv(here("data", "veg.csv"))

# vegetation survey information
metadata <- read_csv(here("data", "vp_veg_metadata.csv"))

# WEEK 5 data

# taxonomic information
taxon_list <- read_csv(here("data", "taxon_list.csv"))

# aquatic invertebrates
aq_ins <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "Aquatic Insects")

# site information
sites <- read_xlsx(here("data", "Aquatic Sampling Data-2026-03-10.xlsx"),
  sheet = "sites")

```

```
# water quality information
water_quality <- read_xlsx(here("data", "Aquatic Sampling
  ↪ Data-2026-03-10.xlsx"),
                           sheet = "Water Quality",
                           na = c("", "n/a", "N/A", "over", "173+"))
```

Class code

```
# WEEK 4

# create a new object from veg
pools_explicit0 <- veg |>
  # filter all occurrences where transect_name contains VP
  filter(str_detect(transect_name, "VP") &
    site == "ncos") |>
  # complete all possible combinations of year, pool, psoc and distance
  complete(year, transect_name, psoc, transect_distance,
    # fill values of percent cover with 0
    fill = list(percent_cover = 0)
  ) |>
  # group by year, pool, and psoc
  group_by(year, transect_name, psoc) |>
  # calculate total percent cover across all quads
  summarize(sum_pc = sum(percent_cover, na.rm = TRUE)) |>
  # ungroup the data frame
  ungroup() |>
  # create a new column called year_pool from year and transect name
  unite("year_pool", year, transect_name, remove = TRUE) |>
  # join with metadata data frame
  left_join(metadata, by = "year_pool") |>
  # calculate mean percent cover from total percent cover/number of quadrats
  mutate(mean_pc = sum_pc/num_quad)

# WEEK 5

# creating new object from sites
ncos_sites <- sites |>
  # filter to only include NCOS sites sampled four times a year
  filter(sector == "NCOS" & sampling_frequency == "Quarterly")
```

```

# creating new clean object from taxon_list
taxon_list_clean <- taxon_list |>
  # converting taxon name to snake case (no spaces or capital letters,
  # only underscores)
  mutate(verbatim_name = to_any_case(verbatim_name, case = "snake"))

# creating new clean object from water quality
water_quality_clean <- water_quality |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites in ncos_sites
  semi_join(ncos_sites, by = "site") |>
  # create new column to join with later data frames
  unite("date_site", date, site, remove = TRUE) |>
  # group by date_site column
  group_by(date_site) |>
  # calculate median pH, salinity, DO
  summarize(
    med_ph = median(p_h, na.rm = TRUE),
    med_sal = median(salinity_ppt, na.rm = TRUE),
    med_do = median(dissolved_oxygen_mg_l, na.rm = TRUE)) |>
  # ungroup data frame
  ungroup() |>
  # filter out salinity outliers
  filter(date_site != "2022-08-12_NVBR")

# creating new clean object from aquatic inverts
aq_ins_clean <- aq_ins |>
  # clean column names
  clean_names() |>
  # filtering join: only include sites occurring in ncos_sites
  semi_join(ncos_sites, by = c("site")) |>
  # renaming columns
  rename(date = date_on_vial) |>
  # select columns of interest
  select(
    date, sample_type, site,
    ostracod,
    copepod,
    hemiptera_corixidae_boatman,
    cladocera,
    nematode,
    diptera_ceratopogonidae,
    annelida_oligochaete,
    diptera_chironomid,
  )

```

```

    annelida_polychaete) |>
# filter to only include "filtered beaker" samples
filter(sample_type %in% c("FB 250", "FB250")) |>
# convert the data frame to long format
pivot_longer(cols = ostracod:annelida_polychaete,
              names_to = "taxon",
              values_to = "abundance") |>
# group by date, site, taxon
group_by(date, site, taxon) |>
# calculate average abundance per liter (sum abundance divided by 7.5
  ↪ liters)
summarize(ave_lit = sum(abundance, na.rm = TRUE)/7.5) |>
# ungroup data frame
ungroup() |>
# create a new column called `date_site` to join with water quality
unite("date_site", date, site, remove = FALSE) |>
# join with water quality
left_join(water_quality_clean, by = c("date_site")) |>
# join with taxon list
left_join(taxon_list_clean, by = c("taxon" = "verbatim_name")) |>
# add full names of sites
mutate(site_full = case_when(
  site == "NVBR" ~ "North of Venoco Bridge",
  site == "NEC" ~ "South of Venoco Bridge",
  site == "NMC" ~ "Main Channel",
  site == "NPB" ~ "South of Phelps Bridge",
  site == "NPB1" ~ "North of Phelps Bridge",
  site == "NPB2" ~ "Phelps Road",
  site == "NWP" ~ "West Pond",
  site == "NDC" ~ "Devereux Creek"
)) |>
# setting factor levels for site
mutate(site_full = fct_reorder(site_full, med_sal, .fun = "median", .na_rm
  ↪ = TRUE),
       site_full = fct_rev(site_full))

```

1. Explain your inspiration

The visualization I chose for my inspiration was [Steven Ponce's "Evolution of MTA Art Materials \(1980-2020\)"](#). I am trying to find the relationship between vegetation and its explanatory variables as well as invertebrate abundance and its explanatory variables. Both datasets have

observations taken from multiple locations, so it makes sense to use this visualization as it allows for multiple panels to be plotted. Additionally, the background lines being subsetted into each panel can help capture the variation of richness of species/abundance of each taxon between each site.

The variables from the veg data set that I will use are: - `transect_name` - `cover_category` - `psoc` - `year` I want to visualize each pool's richness over time with line geometries (representing species richness) to connect them. Colors will vary from each panel based on pool, and the background lines will be grey like Steven Ponce's visual.

The variables from the invertebrate data set that I will use are: - `ave_lit` - `site_full` - `taxon` I want to visualize each site's abundance in different aquatic invertebrates using a log +1 transformation, jitter points for individual observations, and summary points for the median of the average abundance points. Jitter points will be colored a different shade of blue for each site, and the statistic point will be dark blue.

2. Plan your figure

[Visualization plan](#) (please see README)

3. Code your figure

Vegetation

Vegetation code

```
# create new object from veg
native_spp <- veg |>
# filter only native observations
  filter(cover_category == "NATIVE COVER") |>
# pull to extract data frame column as a vector
  pull(psoc)

# random sample
sample(native_spp, 10)
```

```
[1] "Hordeum_brachyantherum_brachy." "Baccharis_pilularis"
[3] "Grindelia_camporum"           "Frankenia_salina"
```

```
[5] "Arthrocnemum_subterminale"      "Extriplex_californica"
[7] "Epilobium_brachycarpum"         "Eleocharis_macrostachya"
[9] "Distichlis_littoralis"          "Heterotheca_grandiflora"
```

```
# create new object from pools_explicit0
species_rich <- pools_explicit0 |>
# group by year and pool
group_by(year, transect_name) |>
# filter data frame with vector \nto get native species,
# also filter percent cover to identify presence of species
filter(psoc %in% native_spp, mean_pc > 0) |>
# species richness calculation
summarize(spp_rich = n_distinct(psoc)) |>
# change pool names to be more understandable
mutate(transect_name = case_when(
  transect_name == "VP-01" ~ "Pool 1",
  transect_name == "VP-02" ~ "Pool 2",
  transect_name == "VP-03" ~ "Pool 3",
  transect_name == "VP-04" ~ "Pool 4",
  transect_name == "VP-05" ~ "Pool 5",
  transect_name == "VP-06" ~ "Pool 6",
  transect_name == "VP-07" ~ "Pool 7",
  transect_name == "VP-08" ~ "Pool 8"
))

# sample 10 rows
slice_sample(species_rich, n = 10)
```

```
# A tibble: 40 x 3
# Groups:   year [5]
   year transect_name spp_rich
  <dbl> <chr>         <int>
1  2021 Pool 1         13
2  2021 Pool 3         19
3  2021 Pool 6         13
4  2021 Pool 7         17
5  2021 Pool 5         12
6  2021 Pool 4         15
7  2021 Pool 8         19
8  2021 Pool 2         14
9  2022 Pool 5          9
10 2022 Pool 7         14
```

```
# i 30 more rows
```

```
# changing years from factors (1-5) to years (2021-2025)
species_rich$year <- rep(2021:2025, each = 8)
```

Vegetation visualization

```
# creating colors for veg visualization
colors <- scale_color_manual(
  values = c(
    "Pool 1" = "#285943",
    "Pool 2" = "#D36582",
    "Pool 3" = "#82204A",
    "Pool 4" = "#74546A",
    "Pool 5" = "#C490D1",
    "Pool 6" = "#FFB7C3",
    "Pool 7" = "#5FB0B7",
    "Pool 8" = "#6C91BF"
  )
)
```

```
# ensuring R reads year column as numeric, not str
species_rich$year <- as.numeric(species_rich$year)
```

```
# creating new ggplot layer, adding data
p <- ggplot(data = species_rich,
  aes(x = year,
    y = spp_rich,
    colour = transect_name,
    fill = transect_name
  )) +
```

```
# Geoms
geom_line(size = 1.7,
  alpha = 0.9) +
# gghighlight to add background lines
gghighlight(
  use_direct_label = FALSE,
  unhighlighted_params = list(
    linewidth = 0.7,
    alpha = 0.75,
```



```

    color = "gray60"
  )
) +
# Adding custom colors
colors +
# manipulating y scales
scale_y_continuous(
  expand = c(0, 0),
  limits = c(0, 30)
) +
# adding labels
labs(
  title = "Species richness of vegetation and abundance of
  ↪ macroinvertebrates at NCOS sites",
  subtitle = "Species richness over time (2021-2025), grouped by vernal
  ↪ pool",
  x = "Year",
  y = "Species richness (n)"
) +
# faceting by pool
facet_wrap(~transect_name,
  scales = "free", ncol = 2
) +
# Theme
theme(
  # adding title customization
  plot.title = element_text(
    face = "bold",
    lineheight = 1.1,
    margin = margin(t = 5, b = 5)
  ),
  # adding subtitle customization
  plot.subtitle = element_text(
    lineheight = 1.2,
    margin = margin(t = 5, b = 20)
  ),
  # adding caption customization
  plot.caption = element_markdown(
    hjust = 0,
    margin = margin(t = 15, b = 5, l = 0),
    lineheight = 1.3
  ),
  # axis customization

```

```

strip.text = element_text(
  size = 12,
  face = "bold",
  color = "gray20",
  margin = margin(t = 5, b = 5)
),
# axis customization
strip.background = element_rect(
  fill = "gray95",
  color = "white",
  linewidth = 0.5
),
# space each panel facet
panel.spacing = unit(1.5, "lines"),
) +
# changing theme from default
theme_bw() +
# removing legend
theme(legend.position = "none")

# showing plot
p

```

Invertebrates

Invertebrate code

```

# new object from aq_ins_clean
log_aq_ins <- aq_ins_clean |>
# mutate to get taxon group -species
mutate(taxon = case_when(
  str_detect(taxon, "annelida") ~ "Annelida",
  str_detect(taxon, "cladocera") ~ "Cladocera",
  str_detect(taxon, "copepod") ~ "Copepod",
  str_detect(taxon, "diptera") ~ "Diptera",
  str_detect(taxon, "hemiptera") ~ "Hemiptera",
  str_detect(taxon, "nematode") ~ "Nematode",
  str_detect(taxon, "ostracod") ~ "Ostracod"
)) |>

```

```
# log + 1 transform mean abundance per liter
mutate(log_ave_lit = log(ave_lit + 1)) |>
arrange(desc(log_ave_lit))

# sample of ten rows
slice_sample(log_aq_ins, n = 10)
```

A tibble: 10 x 24

	date_site <chr>	date <dtm>	site <chr>	taxon <chr>	ave_lit <dbl>	med_ph <dbl>	med_sal <dbl>	med_do <dbl>
1	2021-09-01_NPB2	2021-09-01 00:00:00	NPB2	Dipt~	0	NA	NA	NA
2	2021-11-16_NPB	2021-11-16 00:00:00	NPB	Anne~	0	7.3	2.38	12.1
3	2021-08-25_NPB	2021-08-25 00:00:00	NPB	Clad~	0	7.52	4.57	4.5
4	2023-07-17_NPB	2023-07-17 00:00:00	NPB	Dipt~	0	NA	NA	NA
5	2023-03-03_NPB2	2023-03-03 00:00:00	NPB2	Dipt~	2.67e-1	7.71	1.98	9.59
6	2023-07-17_NPB	2023-07-17 00:00:00	NPB	Dipt~	3.2 e+0	NA	NA	NA
7	2022-08-12_NVBR	2022-08-12 00:00:00	NVBR	Nema~	0	NA	NA	NA
8	2022-08-08_NPB2	2022-08-08 00:00:00	NPB2	Hemi~	0	7.68	2.36	2.43
9	2022-05-12_NPB2	2022-05-12 00:00:00	NPB2	Ostr~	1.20e+3	NA	NA	NA
10	2023-05-04_NMC	2023-05-04 00:00:00	NMC	Clad~	0	8.32	NA	10.3

i 16 more variables: taxonRank <chr>, scientificName <chr>, kingdom <chr>,
Phylum <chr>, Class <chr>, Subclass <chr>, Superorder <chr>, Order <chr>,
Infraorder <chr>, Family <chr>, Subfamily <chr>, Tribe <chr>, Genus <chr>,
comments <chr>, site_full <fct>, log_ave_lit <dbl>

Invertebrate visualization

```
# creating colors object for ggplot
colors2 <- scale_fill_manual(
  values = c(
    "Devereux Creek" = "#07beb8",
    "North of Venoco Bridge" = "#3dccc7",
    "South of Phelps Bridge" = "#68d8d6",
    "Phelps Road" = "#9ceaf",
    "South of Venoco Bridge" = "#c4fff9",
    "North of Phelps Bridge" = "#b2f7ef",
    "Main Channel" = "#bee9e8"
  )
)
```

```

p2 <- ggplot(data = log_aq_ins,
             aes(x = taxon,
                 y = log_ave_lit,
                 fill = site_full
                 )) +
  # geometries
  geom_jitter(height = 0,
             shape = 21,
             width = 0.1) +
# adding median points
  stat_summary(geom = "point",
              fun = "median",
              size = 3,
              color = "darkblue") +
  gghighlight(
    use_direct_label = FALSE,
    unhighlighted_params = list(
      linewidth = 0.7,
      alpha = 0.75,
      color = "gray60"
    )
  ) +
# Scales
  colors2 +
  scale_y_continuous(
    expand = c(0, 0),
    limits = c(-0.3, 7)
  ) +
# adding labels
  labs(
    subtitle = "Median Aquatic invertebrate abundance/L at NCOS sites
    ↪ (2020-2024)\nMedian point shown in dark blue",
    caption = "Data derived from ENV5193DD course, UCSB
    Visualization credit Steven Ponce (https://github.com/poncest)",
    x = "Taxon",
    y = "log +1 transformed average abundance/L"
  ) +
# faceting by site
  facet_wrap(~site_full,
            scales = "free", ncol = 2
  ) +
# wrapping x-axis text
  scale_x_discrete(labels = label_wrap(width = 10)) +

```

```

# Theme
theme(
  # bold title
  plot.title = element_text(
    face = "bold",
    lineheight = 1.1,
    margin = margin(t = 5, b = 5)
  ),
  # adding subtitle
  plot.subtitle = element_text(
    lineheight = 1.2,
    margin = margin(t = 5, b = 20)
  ),
  # adding caption
  plot.caption = element_markdown(
    hjust = 0,
    margin = margin(t = 15, b = 5, l = 0),
    lineheight = 1.3
  ),
  # axis text
  strip.text = element_text(
    size = 12,
    face = "bold",
    color = "gray20",
    margin = margin(t = 5, b = 5)
  ),
  # axis text
  strip.background = element_rect(
    fill = "gray95",
    color = "white",
    linewidth = 0.5
  ),
  # space each panel facet
  panel.spacing = unit(1.5, "lines"),
) +
# changing plot theme from default
theme_bw() +
# removing legend
theme(legend.position = "none")

# showing plot
p2

```

Putting it all together

```
# stacking plots on top of each other  
p / p2
```

4. Describe your process

It was difficult to decide which visualization would best represent the story I was trying to tell for vegetation and invertebrate abundance. I originally chose a different visualization (circle graph), but realized this would look too messy as I was using two different data sets. After selecting the visual, the process was essentially pasting the code into my qmd, looking over each portion to understand what each function does, and replacing it with my own code.

Sometimes, the code would not read because the geometries that Ponce used for his data were different from mine. I would need to go back and identify what portions of the code was causing an error, and then figure out what customization would be best for my data. Some of the packages from his visual would not read in correctly as well, so I did take out some of the code that I felt was unnecessary for my plots anyways.

I ended up going with the “Evolution of MTA Art Materials” because I thought it would be interesting to the other panels data subsetting (as grey lines) onto each site to easily compare data without having to look over at the other faceted panels. This visualization differs from, for example, the figure we created with `stats_summary` in week 5 for aquatic invertebrate abundance, because it is able to show the other jitter points from other sites, while having the main points stand out in color.